MINING DEALER BEST PRACTICE

Prediction of Failure of Starting Motors on D11R Machines

Maintenance and Repair | Application | Component Life Management | Component Renewal (CRC) | MARC Management

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1401-1.02-1330

1.0 Introduction

This Best Practice provides an overview of one strategy in order to predict failures in starting motors of 3508B engines used on D11R machines. These faults widely impact the reliability and availability of the machine.

These failures are related to internal problems of the component or external problems in the electrical starting system. In both cases, these events require maintenance intervention not programmed, increasing the operating costs due to unexpected down of the machine, and the maintenance costs due to investment of resources, manpower, spares, etc.

2.0 Best Practice Description

One of most important issues in the maintenance of electrical systems is the preventive maintenance of the starting system. This is managed only through corrective approach, which can have negative effects in electrical starting system performance and can cause electrical components to not achieve the expected life, especially the starting motors (pre-lube and electrical motor). This preventive maintenance process includes several tasks to provide a reliable way to predict future failures in the starting motors.

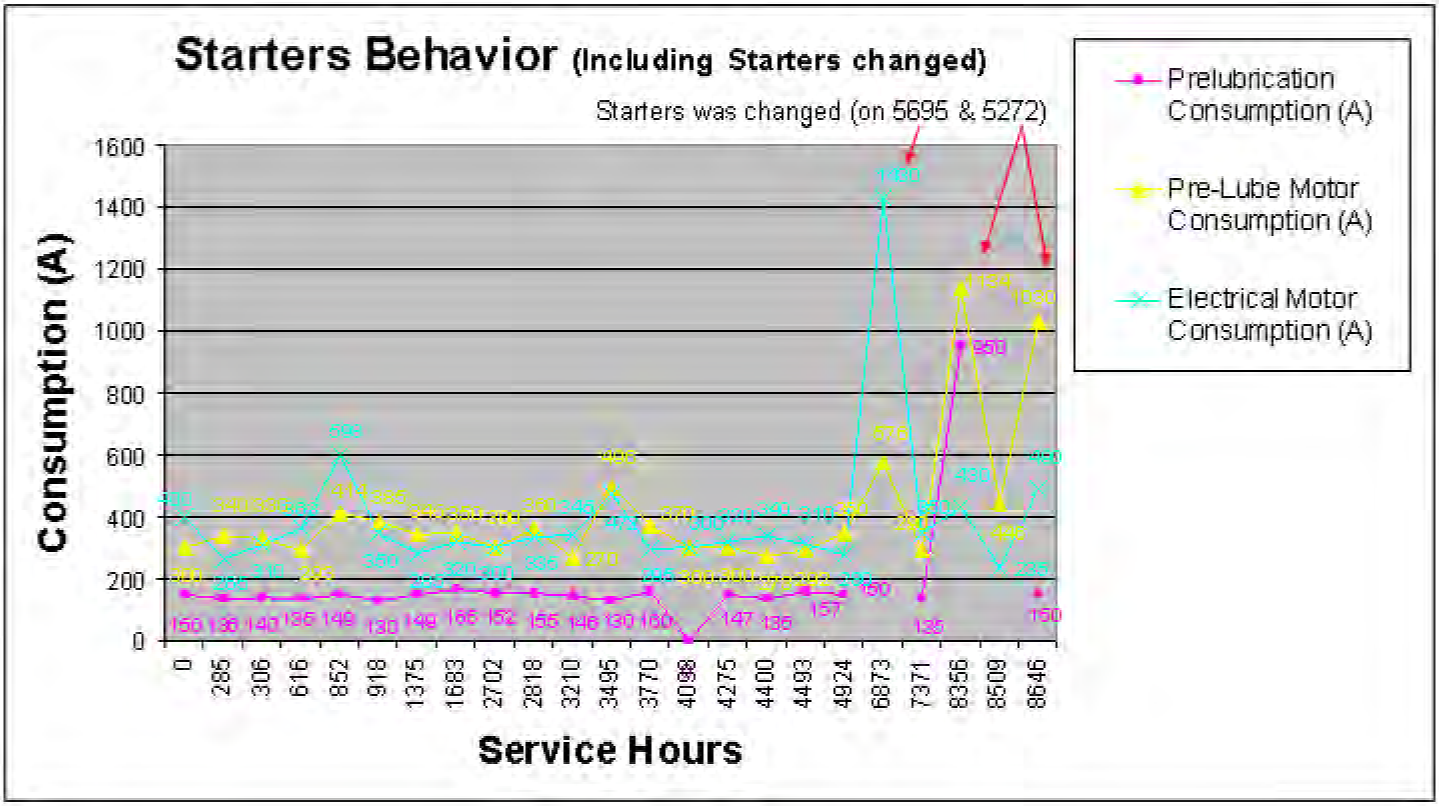
3.0 Implementation Steps

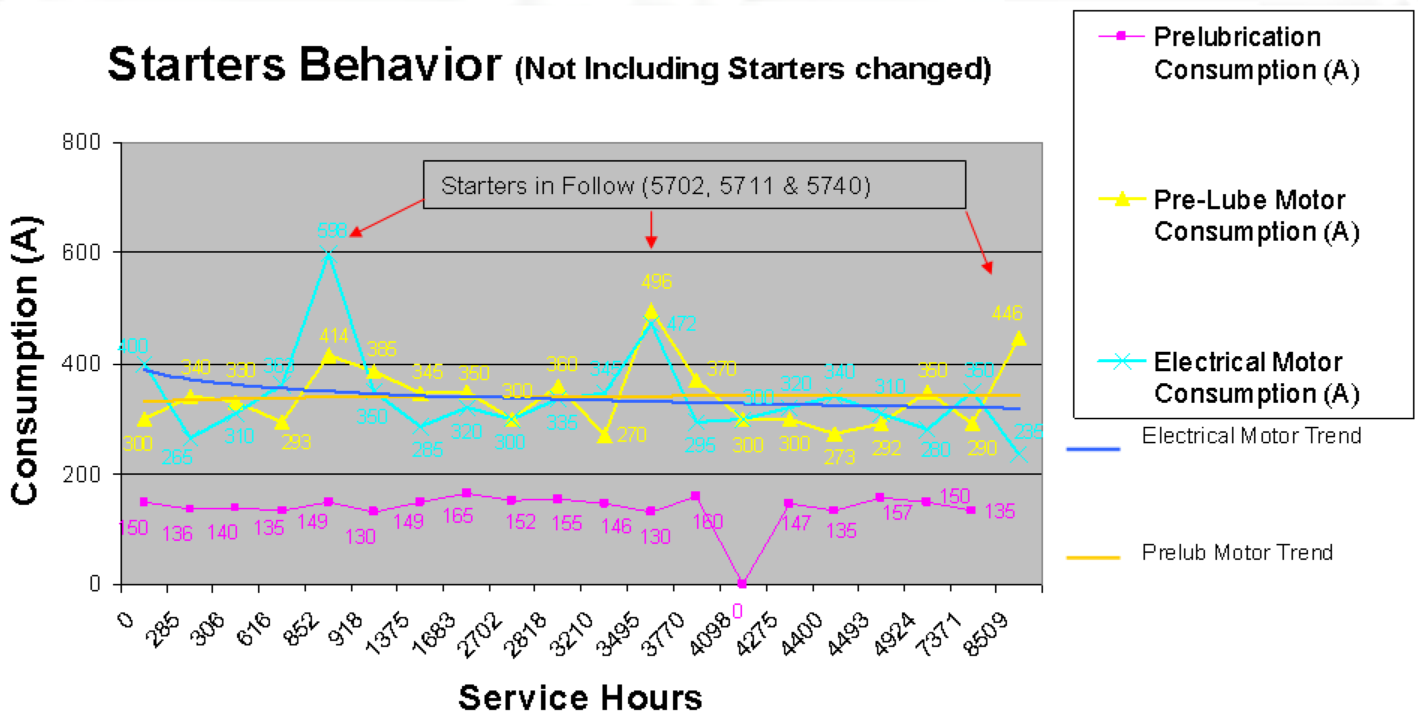
Due to repetitive reports of starting problems which had root cause in the starting motors and there was no information how to predict this failure, an improvement program was conceived in order to identify a mode before failure to avoid unexpected machine downtime.

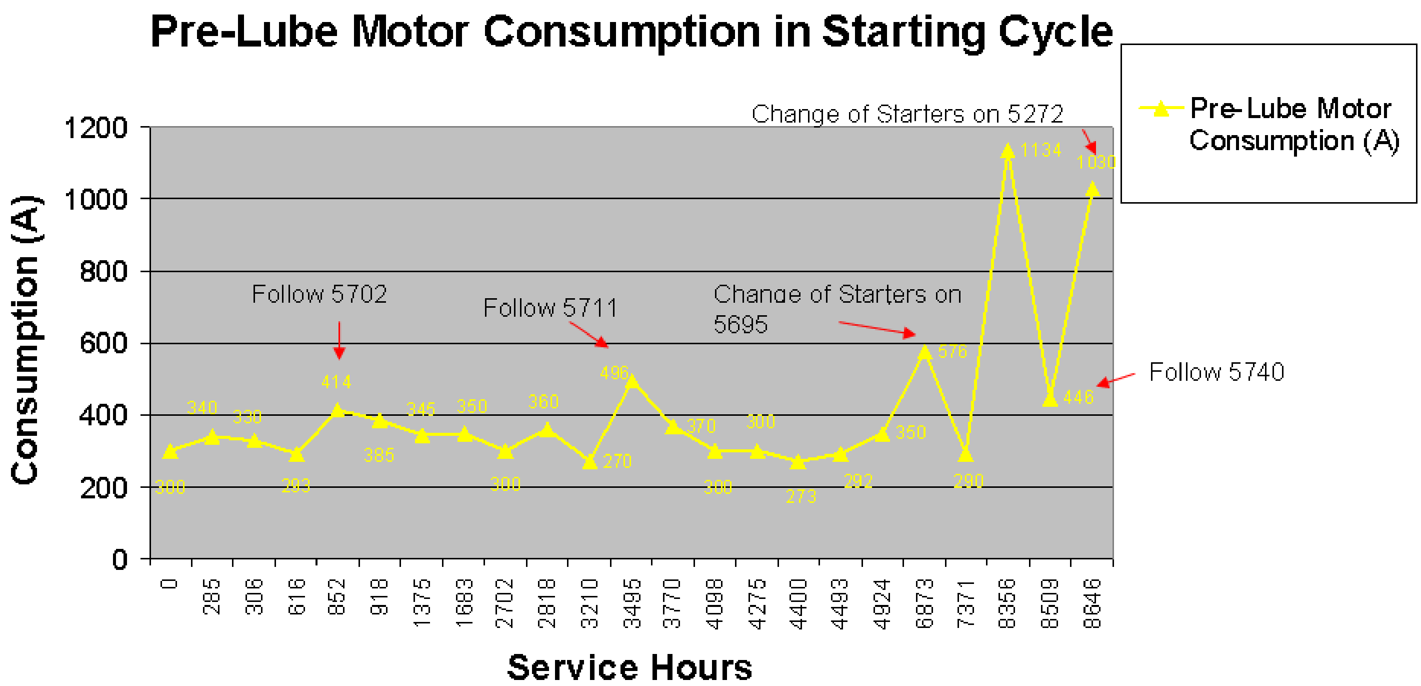
As a result of this program, a procedure was generated in which it was possible to establish a range of energy consumption (in Amps) during the ignition process.

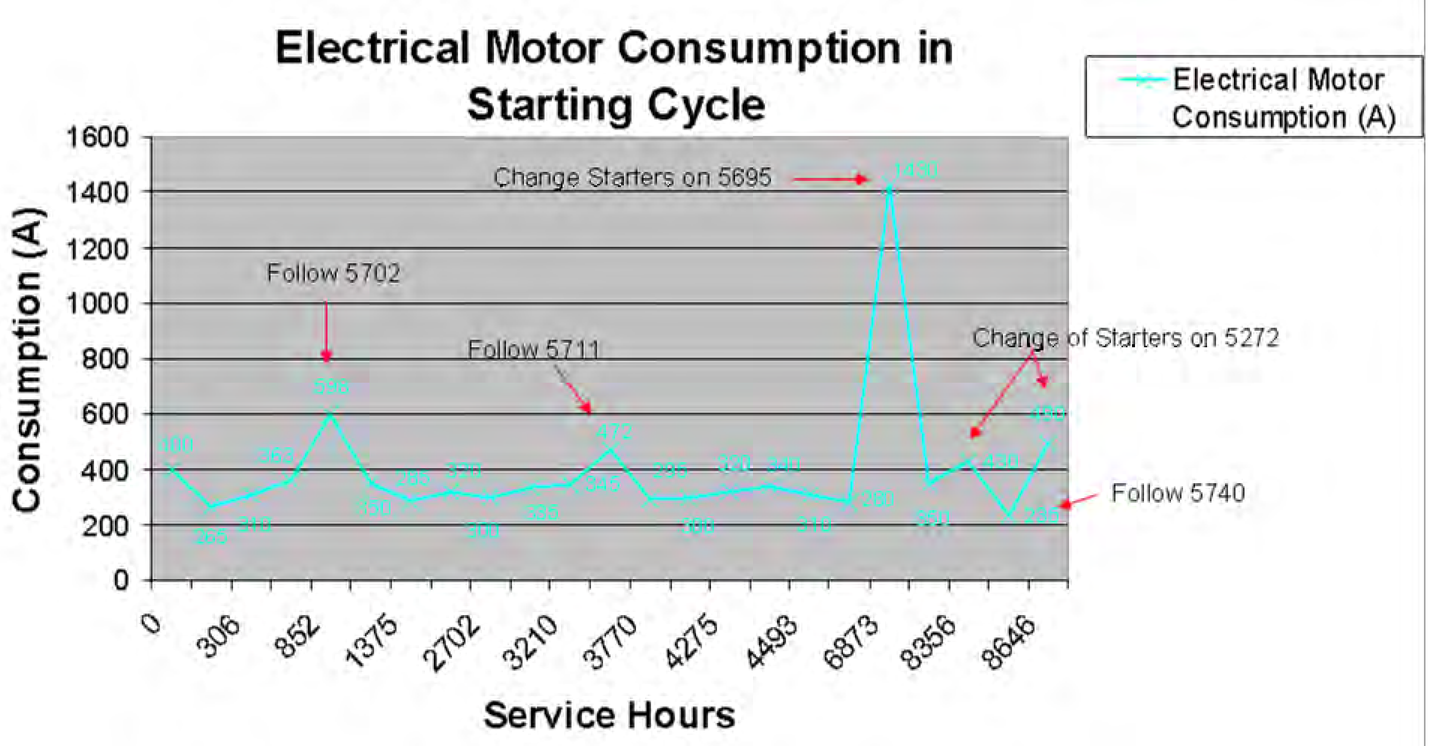
The first step was to develop a form of the procedure in which one may take the information in a logical sequence of execution. As a result, the attached procedure (see 6.0 supporting attachments) was generated.

After we developed the procedure, the second step was based on the data collection to establish a safe range according to the trends in each case. Thus, these data were plotted and analyzed finding the trends and safe operating ranges for the pre-lube motor and electrical motor consumption in starting cycle, and Pre-lubrication consumption. The following graphs show these trends and values.


Starters Behavior (Including Starters changed)
Starters was changed (on 5695 & 5272)
Consumption (A)
Service Hours
Prelubrication Consumption (A)
Pre-Lube Motor Consumption (A)
Electrical Motor Consumption (A)
0
200
400
600
800
1000
1200
1400
1600
0 285 306 616 852 918 1375 1683 2702 2818 3210 3495 3770 4098 4275 4400 4493 4924 6873 7371 8356 8509 8646


Starters Behavior (Not Including Starters changed)
Starters in Follow (5702, 5711 & 5740)
Consumption (A)
Service Hours
Prelubrication Consumption (A)
Pre-Lube Motor Consumption (A)
Electrical Motor Consumption (A)
Electrical Motor Trend
Prelub Motor Trend
0
200
400
600
800
0 285 306 616 852 918 1375 1683 2702 2818 3210 3495 3770 4098 4275 4400 4493 4924 7371 8509





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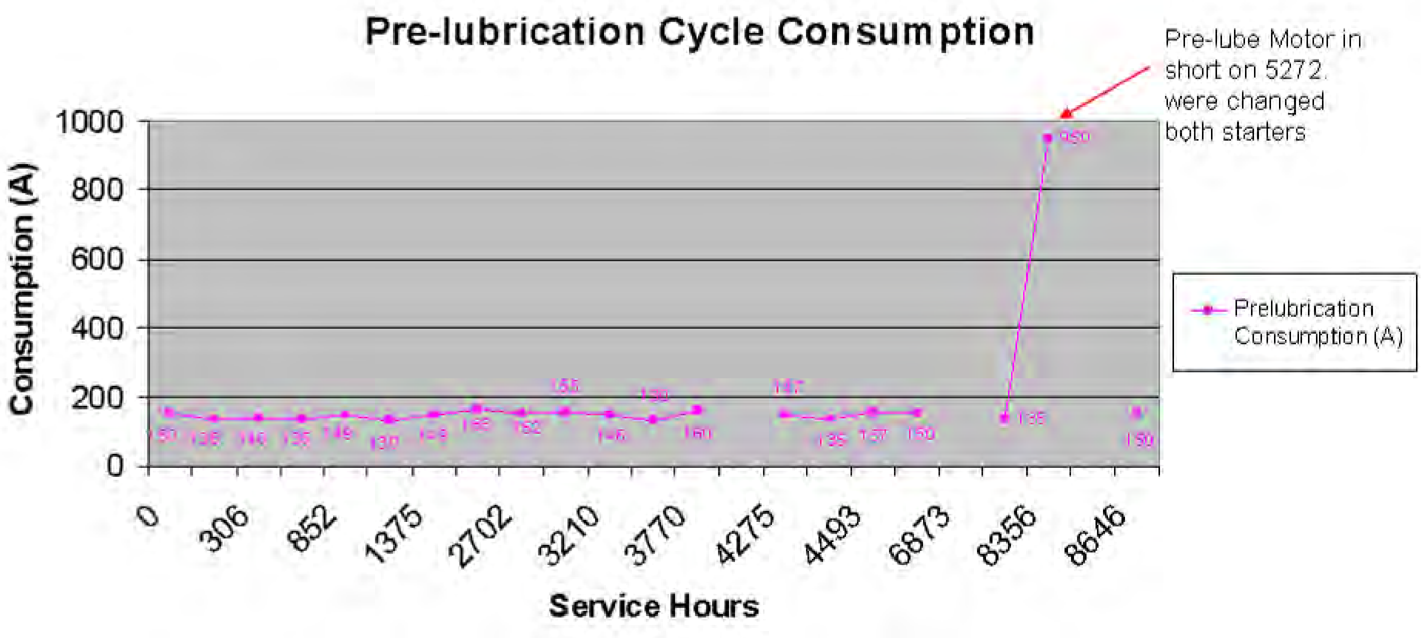


As result of the analysis, we stated:

- The Consumption of pre-lubrication cycle should be between 130 and 160A.
- The Consumption of both starters in starting cycle should be up to 450A.
- The test was included in all Preventive Maintenance service work, every 280 hrs.

SUMMARY CHART			
Description	**Condition**	**Consumption (Amp)**	**Action Required**
Starting Motors (each one)	Starting Cycle	Up to 450	No action Required. Execute Preventive Maintenance
		450 - 550	Follow Up. Check Starting system components for anomalies
		550 - Up	Change
Pre-Lube Motor	Pre-Lubrication Cycle	130-160	No action Required. Execute Preventive Maintenance
		160 - Up	Change

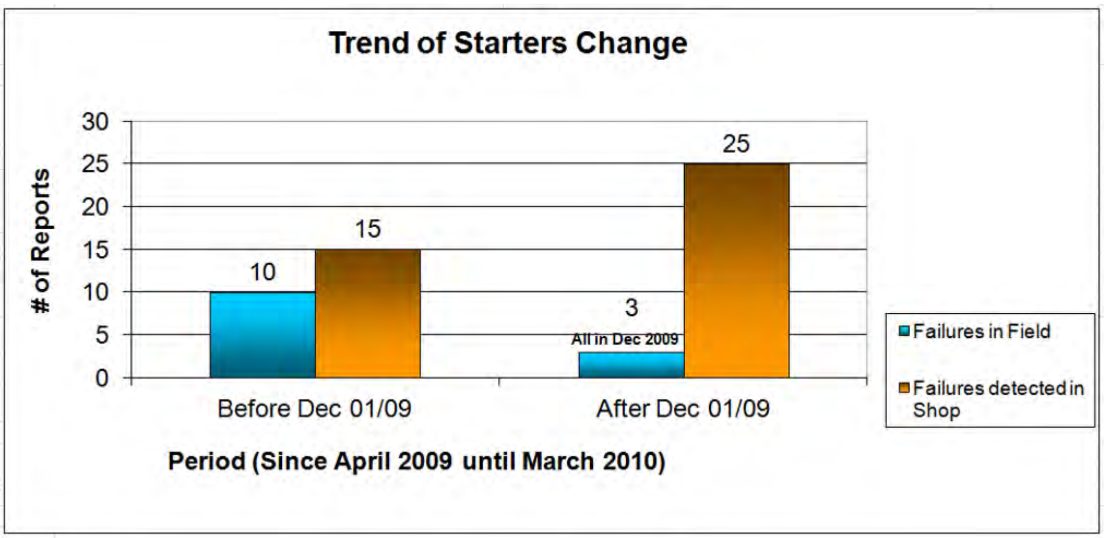
This Best Practice is applied to the following starter motors in engine 3508B

ENGINE	STARTING MOTOR	STARTING MOTOR PRELUBRICATION
3508B	6V-0889 6V-0513 338-3455 338-3456	123-8689 123-8686

4.0 Benefits

The benefits acquired of the implementation of this practice are:

- Improvements in the maintenance process of electrical system in general.
- Prediction of failures in the starter motors thus avoiding an unscheduled stop in the machine improving the reliability and availability of the fleet. This is reflected in lower operation and maintenance costs.



- Saving in man power (technicians and tools transporting costs to field are not included) and saving in reduction of machine down time in field equivalent to 54%; total Saving $91,490 USD/year. (see chart)

Benefits in Manpower Savings and Reliability			
Description	Failures in Field	Failures detected at Shop	% Reduction
Machine Down time (Average per Call)	36	12	67%
Man Power per Call (2 Technicians)	72	24	67%
Cost of 1 Drummond Technician Hour (USD)	$9.5	$9.5	
Cost Man Power per Call (USD)	$684	$228	67%
Calls in 2009	13	15	100 % of Reduction of Failures in Filed (2009 vs 2010)
Calls in 2010	0	25*	
Overall Cost Man power in 2009 (USD)	$12,278		54%
Overall Cost Man power in 2010 (USD)	$5,700		
Savings in Man Power (USD)	$6,578		

* These failures were detected simultaneously to PM jobs and not represent a call

Benefits in Costs of Machine Operation		
Description	Failures in Field	Failures detected at Shop
Machine Down time (Average per Call)	36	12
Calls in 2009	13	15
Calls in 2010	0	25
Total Machine Down Time in 2009	648	
Total Machine Down Time in 2010	300	
Machine Hours Reduction	348	
Cost/Operation Hrs of Machine (USD/Hr)	$244	
Savings in Machine Operation (USD)	$84,912	

5.0 Resources Required

The estimated investment related to this project is approximately US $1,255. This is the cost of a Clamp-On Multimeter P/N 329-9303 which will be invested only once. Besides 2 man hrs per machine of preventive maintenance that can be executed in parallel to others major tasks to avoid additional downtime.

6.0 Supporting Attachments / References

- Procedure of measurement

Drummond LTD	PROCEDIMIENTO DE MEDICIÓN DE AMPERAJE DE MOTORES DE ARRANQUE EN MOTORES C32	

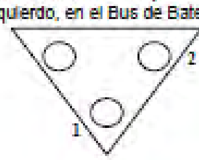
EQUIPO:________________ SMU:_______ TEC:____________________

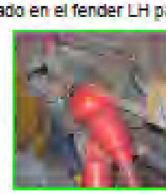
FECHA :________________ GRUPO:____ SUP:____________________

Objetivo: La siguiente prueba tiene por objetivo predecir las fallas de los motores de arranque.
Procedimiento:

1. Verifique todas las conexiones del sistema de arranque en el equipo.
2. Verifique que las baterías se encuentran en normal operación y con la carga correspondiente.
3. Coloque Shut-Down al Motor.
4. Utilice una pinza Volti-Amperimétrica P/N 227-4324. Colóquela en la escala A ⎓ (Corriente DC).

5. Coloque la pinza en los cables que van hacia cada uno de los arranques, ya sea el Derecho o el Izquierdo, en el Bus de Batería ubicado en el fender LH parte interna:





Forma correcta de Medir en los cables hacia los motores de arranque

6. Gire la llave hacia la posición de encendido durante unos segundos y tome las mediciones. Regístrelas en la Tabla 1.

TABLA 1

Ciclo de Arranque	
Consumo del Motor RH (A)	**Consumo del Motor LH (A)**

7. Desconecte el Shut Down.
8. Cerciórese que la máquina encienda correctamente.

7.0 Related Best Practices

- None.

8.0 Acknowledgements

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